MOTOR: Musculoskeletal Optimisation Through Open Robotics Simulation

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Abstract

Ground reaction forces, joint moments and muscle forces say more about how a body is loaded than joint angles do, but they still come from laboratory equipment or from slow musculoskeletal software that is tied to one kind of input. We present MOTOR, an open musculoskeletal solver that takes a moving body from any source, given as a skin mesh, a rotation matrix per segment and the joint centres, and returns the movement of a full-body musculoskeletal model (MyoFullBody, 416 muscles) together with the forces that move it. The solver drives the model directly with one fit per frame: the pose that comes closest to the input body while every joint stays within its anatomical limits. The segment rotations are the main target, each coupled to the group of bones that covers the same piece of body; about a hundred virtual markers on anatomical landmarks place the joint centres and the lengths; and the bones are held inside the body's skin. The fit is solved in three levels, each fixed for the next: once over many different bodies, once per person and once per frame. The whole trial is then solved as one dynamics-consistent optimisation on a single GPU, with Newton and MuJoCo Warp. Because MOTOR is a solver and not a pipeline, it applies to motion from video estimators, from motion generators such as Kimodo, and from motion capture alike. We validate it the way most users will meet it, as a monocular pipeline: one ordinary smartphone video, an off-the-shelf body estimator (GEM-X), then MOTOR, on the OpenCap LabValidation cohort against marker-based motion capture, force plates and EMG, with five subjects held out from every choice of setting.

Gap: The headline numbers go here: MOTOR's pooled joint angle error on the matched OpenCap Monocular envelope, its vertical and anterior ground reaction force error, its joint moment error against OpenCap's, and its time per trial on one GPU against OpenCap's recorded solves, all from the whole-trial solver on the 720 single-camera inputs with the held-out subjects 7 to 11 apart. They wait on the solver's build (#98, steps 4 to 6) and its full run.

MOTOR is open source under the MIT licence.

1 Introduction

What a movement does to the body is carried by forces. The ground reaction force under each foot, the net moment at each joint and the force in each muscle describe how a knee is loaded in a squat, how hard a calf works in push-off, or how one leg spares the other after an injury; joint angles describe only the shape of the movement that produced them. These forces are measured in a gait laboratory with force plates and marker-based motion capture, and the muscle forces, which no instrument measures directly, are estimated from that data by a musculoskeletal model in software such as OpenSim [1–3] or its optimal-control extension Moco [4]. The laboratory, its operators and its solve times keep these quantities out of most clinics, sports settings and studies [5].

OpenCap moved this analysis to smartphones [6]. From two or more synchronised phone videos it estimates keypoints, fits a scaled OpenSim model by inverse kinematics on virtual markers, and

solves a muscle-driven tracking simulation for the ground reaction forces, joint moments and muscle activations. On its LabValidation cohort it recovers joint angles within a few degrees of motion capture, and kinetics close enough to tell clinically relevant patterns apart, such as an asymmetric squat. OpenCap Monocular brings the kinematics and dynamics to a single video [7]. Its dynamic solves, however, take tens of minutes per trial on the CPU, and the pipeline is built around its own keypoints and model.

Others have moved musculoskeletal fitting onto the GPU and toward a body model. Cotton fitted biomechanical models to video as differentiable models in MuJoCo [8, 9]; the Portable Biomechanics Laboratory fits one body scale jointly across all of a person's trials from a handheld phone [10]; KinTwin recovers joint torques, muscle activations and ground reaction forces from markerless capture [11]; and the Biomechanical 3D Body fits a person's constants over frames sampled from all of their recordings and weighs its landmarks by their own residuals [12], two ideas MOTOR uses. SKEL re-rigs a parametric body model with a biomechanical skeleton that sits inside its skin [13], and AddBiomechanics automates model scaling, inverse kinematics and inverse dynamics from motion capture [14]. On the simulation side, MyoSuite provides MuJoCo musculoskeletal bodies [15], MuJoCo Warp runs MuJoCo on the GPU [16], and Newton builds a GPU physics engine on Warp around it [17, 18]; MuscleMimic [19] and MS-Human-700 [20] train controllers for bodies of 400 to 700 muscles there, and static optimisation on MuJoCo models has run on the CPU [21, 22].

Meanwhile, parametric body models have become the common output of human motion work. Video estimators such as GEM-X [23, 24] recover a body model's mesh and per-segment rotations from one video, motion generators such as Kimodo [25] produce the same rotations from text, and the SOMA convention [26] unifies several body models under one skeleton. Such a body already carries most of what a musculoskeletal fit needs: a rotation per segment, joint centres and a skin. Most musculoskeletal pipelines use little of it. Inverse kinematics on virtual markers keeps only points, so each segment's rotation about its own long axis must be recovered from markers a few centimetres apart, and nothing keeps the fitted bones inside the body they came from.

This paper contributes a solver for that step: from a moving body, given as a skin mesh, segment rotations and joint centres, to the dynamics of a full-body musculoskeletal model. What is new is the combination of four elements. First, a per-frame fit that takes the segment rotations as its main target, each coupled to the group of bones in MyoFullBody that covers the same piece of body (one spine segment against several vertebrae, for example), with about a hundred virtual markers on anatomical landmarks beside them to place the joint centres and lengths, the bones held inside the input's skin, and the joint limits and couplings held exactly. Second, a three-level solution of that fit: once over many different bodies for each landmark's place and weight, checked on bodies not in that tuning; once per person for the person's constants from all of their frames together; and once per frame for the pose alone. Third, the whole trial solved as one dynamics-consistent optimisation on a single GPU with Newton and MuJoCo Warp, with the per-frame fit as its warm start. Fourth, a judgement of the fit on how well the model follows the body (joint positions, segment rotations, bones inside the skin), never on the marker distances the solver itself pulls on. To our knowledge MOTOR is the first open solver from a body's motion to joint moments and muscle forces on Newton and MuJoCo Warp.

We validate MOTOR as a monocular pipeline, the way most of its users will meet it: one smartphone video, GEM-X, then MOTOR, on the OpenCap LabValidation cohort [6], against marker-based inverse kinematics, force plates and EMG, and compared with OpenCap on its own cohort and reference. The same solver runs inside SCORE, an open environment synthesis framework, on Kimodo's generated motion; other estimators and motion capture fit the same input, and their validation is left to future work.

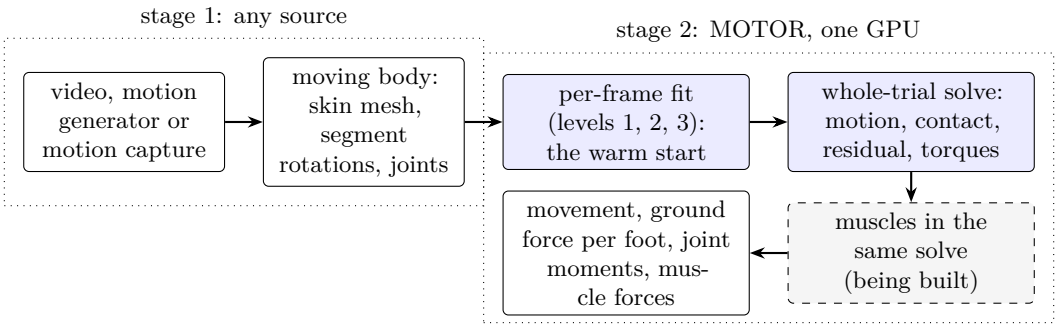


Figure 1: The pipeline. Stage 1 supplies a moving body from any source; in the validation it is GEM-X on one smartphone video. Stage 2, MOTOR, fits MyoFullBody to that body frame by frame and then solves the whole trial at once on one GPU. The dashed step, muscles inside the whole-trial solve, is being built (Section 2.6).

2 Methods

2.1 Pipeline overview

The pipeline has two stages (Figure 1). The first turns sensor data, or a generator's output, into a moving body: per frame a skin mesh, a rotation matrix per segment of a kinematic chain, and that chain's joint centres. The second, MOTOR, turns that body into the movement of a musculoskeletal model and the forces that move it. This paper is about the second stage. The first is a supplier of inputs; any source that gives the three parts can serve in that role.

The validated instantiation uses GEM-X [23, 24] on one smartphone video. GEM-X gives a SOMA body [26] per frame at 30 Hz: its identity (shape), root translation and global segment rotations. SOMA is the truth throughout: MOTOR never changes its shape or pose. GEM-X is given the camera's field of view that MoGe-2 [27] estimates from the same video, and its own contact post-process holds the static feet. GEM-X's world has no floor, so the floor is seen instead: a plane fitted to MoGe-2's floor points beside the person, the clip turned onto it in one rigid turn and lowered until its feet's lowest point meets it.

2.2 The musculoskeletal model

The model is MyoFullBody from MyoSuite [15], its arm and hand from MoBL-ARMS [28, 29]: 416 muscles, 86 bones and 132 joints, of which one is the free root, 36 are coupled to another joint by a polynomial (the knee's and the shoulder girdle's, for example) and 95 are free to move on their own. Its bone groups are sized to the person's SOMA body, each group's length taken from its SOMA segment, joint to joint; MOTOR does not solve for any length. Its masses are SOMA's segment volumes spread over the subject's measured mass. Inputs SOMA does not give are recorded as assumed.

2.3 Segments, bone groups and landmarks

Segment rotations and bone groups. Each SOMA segment is coupled to a *bone group*: the bones of MyoFullBody that cover the same piece of body, with one of them, the group's frame body, carrying the segment's rotation. A spine segment is coupled to the vertebrae of its height, paired by their share of the spine's height; the hand to the bones of the wrist and hand, the lunate carrying its rotation. A segment drives one group, one to one; where an earlier group already rides the same frame body (SOMA's two neck segments on MyoFullBody's one rigid neck), the second drives none

and is still judged. How a segment lines up with its group is read once from the two models at rest, along the line between the group's two end joints, and stored as a static calibration table: a fixed rotation per group, never fitted to data and never per person.

Virtual markers on anatomical landmarks. Beside the rotations the fit takes virtual markers on anatomical landmarks, each defined by its anatomy following the ISB recommendations [30, 31]: the acromion, the humeral epicondyles, the anterior and posterior superior iliac spines, the malleoli, the metatarsal heads and so on. Each landmark sits on a fixed point of the SOMA mesh, given as a face and barycentric weights, so it moves with the person's skin, and on the matching point of a MyoFullBody bone, offset from it by a link toward the skin. The set holds 102 landmarks from a landmark study on the two models; the solver takes 84 of them, each with a trust. No licensed dataset's marker list is used.

Joint centres as segment ends. SOMA's rig joints mark where each bone group starts and ends: the hips, knees, ankles, shoulders, elbows, wrists and the joints between spine segments. Each such end is one more marker, at MyoFullBody's own joint centre there, aimed at SOMA's joint. The joints inside a group (the lumbar levels, the shoulder blade, the wrist bones, the joints of the inner foot) are not aimed at anything; the landmarks and the model's couplings place them. No joint gets a special case.

The skin. The bones belong inside the body they came from. This idea is the one SKEL builds into a body model, a biomechanical skeleton placed inside the skin of SMPL [13]; MOTOR uses the input's own skin as a target instead. Each frame's SOMA skin is posed, and every kept vertex of every bone mesh is measured against it: its offset to the closest point of the skin, zero when a generalised winding number puts it inside.

2.4 The per-frame fit

For one frame, with the person's constants fixed, the fit finds MyoFullBody's coordinates q that come closest to the input body while every joint stays within its anatomical limits:

$$\min_{q} \frac{1}{N_g}\sum_{g}\left\|\log\!\left(\hat{R}_g^{\top} R_g(q)\right)\right\|^2 \;+\; \frac{\lambda_m}{\sum_m w_m}\sum_{m} w_m \left\|p_m(q)-\hat{x}_m\right\|^2 \;+\; \frac{\lambda_s}{N_b}\sum_{b}\left\|d_b(q)\right\|^2 \tag{1}$$
$$\text{s.t. } q_{\text{follower}} = f(q_{\text{leader}}) \text{ for every coupling}, \qquad q_{\min} \le q_{\text{leader}} \le q_{\max}.$$

The first term is the main one: for each of the N_g bone groups, the angle between the group's calibration frame $R_g(q)$ and its SOMA segment's rotation $\hat{R}_g$ carried through the static table. The second pulls each marker m (the landmarks and the segment ends) at its point $p_m(q)$ on the model toward its point $\hat{x}_m$ on the SOMA body, weighed by its trust w_m. The third holds the N_b bone vertices inside the skin, d_b being a vertex's offset outside it. Each goal is normalised by its own count, as OpenSim's inverse kinematics weighs its goals [2]; the turns are the main term, the markers' goal λ_m is set by level 1 (2.1169 on main), and the skin goal λ_s is 10^3, chosen in a sweep that traded the bones outside the skin against the turns. The couplings are MyoFullBody's own polynomials, held as equality constraints; the limits are the published ranges of the leading joints, held as bounds. The fingers are held at SOMA's rest.

The reference solver is mink [32], a differential inverse-kinematics library on MuJoCo: per step one quadratic programme with the rotations and places as frame tasks, the couplings as equality constraints, the leading joints' ranges as configuration limits and the fingers frozen, each step judged

in the manner of Levenberg–Marquardt. The skin term enters as three offset rows per bone point, whose Jacobian is the projector off the closest feature of the skin mesh (a face, an edge or a corner). On the GPU the same step runs for every frame at once, as a damped Gauss–Newton step in the reduced coordinates with Warp's tile Cholesky [18]; on every frame of the proto clip it equals mink within 0.0025° in every angle and 0.0042 mm in every site. A frame that does not converge in 300 steps stops the run with an error; it is never kept silently.

2.5 Solving in three levels

The fit has constants that do not change from frame to frame: where each landmark sits and how much it is trusted, and where the joints inside each bone group sit in this person. They are solved in three levels, each held fixed for the levels that follow, and every level is judged on how well the model follows the body, never on the marker distances the solver pulls on.

Level 1, once over many bodies. One bounded least-squares problem tunes every landmark's link (a joint-defining one only across its bone), every landmark's trust (left and right sharing theirs, mirrored), every segment end's shift on MyoFullBody within 40 mm of its start, and the two global weights. Its objective has two terms: each body region's bones' depth outside the SOMA skin, and each segment's turn error against its bone group, each normalised by its own value at the start so that the two weigh alike. Each evaluation runs levels 2 and 3 on every tuning body. The bodies are synthetic SOMA bodies (tall, short, heavy, slim, long- and short-legged) doing seven clips each (a Kimodo walk, a rig walk, arm raises, a reach, a squat, running and sit-to-stand); they are split by body into tuning and check bodies, every clip of a body on one side. The check bodies are scored during the run and never used to choose; a point is accepted when they score within 10% of the tuning bodies. Level 1's gradient is taken through the inner levels by implicit differentiation, with the columns that path does not yet carry by finite differences, and every column is checked against finite differences before a run is trusted.

Level 2, once per person. With level 1 fixed, the person's constants are solved from all of their frames together: 30 frames spread over the person's clips are posed, and one sparse least-squares problem moves each free marker on its body within 3 cm of its link, pulled to it at its own weight, and each joint inside a bone group within its parent body, together with the sampled frames' coordinates [12]. Only the directions the data can tell apart are solved; an inside joint's axis whose column of the problem's Jacobian lies in the span of the others stays where the model has it. The markers on the plantar pads and on the skull never move, so a foot cannot sink through the sole of the skin nor the skull into the head. The skin term is left out of level 2: with it, the problem traded the person's constants for skin, moving inside joints by 40 to 52 mm. The constants are written into the person's model file, so every later solve uses them.

Level 3, per frame. With the person's constants fixed, every frame solves only its pose, by Equation (1): the rotations as the main term, the markers beside them, the bones inside the skin, and the joint limits and couplings as hard constraints.

2.6 The whole-trial solve

The per-frame fit follows the body but knows nothing of forces: it can put a foot where no ground could hold it, and the frames it gives need not be consistent with the forces that would move the

model between them. MOTOR therefore solves the whole trial as one optimisation, with the per-frame fit as its warm start. Its build follows a fixed order, each step measured before the next: (1) the whole-trial movement, (2) inverse dynamics, contact and the pelvis residual, (3) tuning of its physical settings, (4) the muscles, (5) batching across trials, and (6) the move into the stage that SCORE runs. Steps 1 to 3 are on main and described here; steps 4 to 6 are being built.

The movement. Every leading coordinate's path over the trial is one uniform cubic B-spline with one control point per frame, so speeds and accelerations are the spline's own derivatives and there are no kinematic defects to constrain. The root's place is a spline like a joint's; its turn is the fit's turn low-passed and multiplied by the exponential of a body-frame rotation-vector spline. Step 1's cost is the fit's own cost of Equation (1) over every frame at once, plus the spline's jerk, with the leading joints inside their ranges by bounds on their control points. The jerk weight of each coordinate is not tuned: it is the one that puts the solve's smoothing at the 6 Hz cut-off the per-frame chain filters with [5]. Each frame's Gauss–Newton block comes from the GPU fit's own pieces, and the normal equations are banded: each frame's rows touch only its three control points.

Dynamics, contact and the residual. Step 2 (stage A) adds to the same unknowns the dynamics of every frame. MuJoCo Warp, inside Newton, gives the generalised forces $\tau_{\text{needed}}(q, \dot{q}, \ddot{q})$ each frame's motion needs. The ground's force comes from the motion alone, with no force unknowns: every point of the person's skin within 2 cm of its floor is pushed by a smooth spring and damper of the Hunt–Crossley kind with a smoothed Coulomb friction, the form of OpenCap's tracking simulation [6],

$$f_n = k\,\sigma_s(-h)\,\sigma\big(1 - c\,v_n\big), \qquad \mathbf{f}_t = -\mu\,f_n\,\frac{\mathbf{v}_t}{\sqrt{\|\mathbf{v}_t\|^2 + \varepsilon^2}}, \tag{2}$$

with h the point's height over the seen floor, v_n and $\mathbf{v}_t$ its normal and tangential speed, and σ_s and σ smoothed ramps, so that a lifting foot pushes nothing and never pulls. On the leading coordinates, through the couplings' map G,

$$\tau = G^{\top}\big(\tau_{\text{needed}} - J_c^{\top} f_c\big). \tag{3}$$

Its root's six rows are the pelvis residual, the force and moment no part of the body or the ground explains; they are held by an augmented Lagrangian under 2% of body weight and 1% of weight times height, OpenSim's limits [33], on every frame. The joints' rows are the torques, kept small as an effort term. The whole is solved by Levenberg–Marquardt on the banded normal equations; the needed forces' Jacobians come from MuJoCo's mass matrix (exact in the acceleration), the joints' damping (in the speed) and one-sided differences on the GPU (in the place). Before the first round the contact is settled: a body whose lowest skin point sits under its floor at the warm start has its floor lowered by that depth on that frame, so that a seat placed from the protocol under a skin the video sat lower does not push the body with many times its weight.

Settings. The contact's settings are physical: a whole foot's stiffness of 6×10^6 N/m spread over its skin points (OpenCap's six contact spheres per foot at 10^6 N/m each), a dissipation of 2.0 s/m and a friction of 0.8 as in OpenCap's contact spheres, a slip transition ε of 0.05 m/s, and an effort share of 0.1 against the fit's cost at the warm start. Step 3 tuned them on subjects 2 to 5 only, against the force plates with stage A's own contact, and kept these first values (Section 3.4). Nothing is ever fitted to the reference joint angles.

Muscles. Step 4 puts the muscles into the same solve: per person polynomial muscle–tendon lengths, their derivatives the moment arms, fitted to MuJoCo's own tendon lengths, and MuJoCo's muscle force curves, so that the activations are solved together with the motion and the contact rather than distributed afterwards.

> **Gap:** Step 4's formulation as built (the muscle terms, the activation dynamics and how they enter the solve) replaces this box once it is on main and measured (#98).

2.7 The warm start: the per-frame chain

Before the whole-trial solve existed, MOTOR's forces came from a chain of per-frame steps, which now gives the whole solve its starting point. After the fit, the standing feet are put on the floor by moving the whole body up or down, never by changing the pose; a residual reduction in the manner of OpenSim's RRA adjusts the pelvis, the trunk and the masses within a per-frame cap so that the ground can explain the movement; and the forces are split in the two stages of OpenSim's static optimisation [1], first the ground reaction force from the pelvis' six balance equations alone, then the muscle activations with that force fixed. Every step runs every frame at once on the GPU in Warp's deterministic mode, and two runs agree bit for bit. On LabValidation the chain's matched joint angle error is 5.16° (95% CI 4.77 to 5.59°), its vertical ground reaction force error 16.2% of body weight, its joint moment error 2.41% of weight times height, and it takes a median of 66 s per trial on one NVIDIA L4; its pelvis residual stays within both of OpenSim's limits on only 21% of frames. Appendix B gives its full validation. These are the warm start's numbers, not MOTOR's result.

2.8 Validation protocol

We validate on OpenCap LabValidation [6]: nine healthy subjects (2, 3, 4, 5, 7, 8, 9, 10 and 11; subject 6 is not part of the published cohort) in walking, walking with trunk sway, squats, sit-to-stand and drop jumps, recorded with marker-based motion capture, force plates and EMG of 16 lower-limb channels. MOTOR sees only smartphone video, one camera at a time. Its reference standard is the lab's: joint angles from the markers through OpenSim inverse kinematics on the lab's model, ground reaction forces from the plates, joint moments from OpenSim's inverse dynamics remade on the lab's own model and settings, and EMG envelopes for the muscle activations.

Camera input. The video comes from five synchronised iPhones at 60 Hz, one at 0° and pairs at ±45° and ±70°. The 144 non-static trials times five cameras give 720 single-camera inputs, each run on its own. For the matched comparison we use the 45° camera (Cam1) on the non-jumping activities and the fifteen joint angles that have a counterpart in OpenCap Monocular's set, the envelope on which OpenCap Monocular reports its results [7] and on which the author's earlier work reported GEM-X's angles [34]. The full technical validation pools all 720 inputs, every activity and every camera.

Metrics. Video and reference are not synchronised at the millisecond level, so each trial gets one time shift of the reference before scoring. Joint angles are scored by mean absolute error (MAE), centred MAE (cMAE, each trial's constant offset removed, separating tracking from offset), root mean square error and Pearson's r, with a 95% confidence interval from a per-subject bootstrap of 20,000 draws. Ground reaction forces are scored as OpenCap scores them, in % of body weight on three axes with both feet summed, and per foot by the correlation of the vertical force; joint moments

Table 1: The headline comparison. Joint angles on the matched envelope (Cam1, no drop jumps, 15 coordinates); kinetics over all cameras and activities. GRF in % body weight, both feet summed; moments in % weight × height. OpenCap: two cameras, its published figures (OpenCap Monocular's angle MAE: 4.8°). Warm start: the per-frame chain (Section 2.7), not MOTOR's result. Held out: subjects 7–11.

	OpenCap	Warm start (chain)	MOTOR	MOTOR, held out
Joint angle MAE, matched 15 (°)	4.5	5.16	*gap*	*gap*
Joint angle cMAE, matched 15 (°)	–	3.06	*gap*	*gap*
GRF vertical MAE	11.4	16.2	*gap*	*gap*
GRF anterior MAE	3.5	7.4	*gap*	*gap*
GRF medio-lateral MAE	3.8	3.0	*gap*	*gap*
Joint moments, OpenCap's 15, MAE	1.2	2.41	*gap*	*gap*
Early-stance knee adduction peak, left	0.30	1.51	*gap*	*gap*
Vasti symmetry AUC against the EMG label	0.83	0.66	*gap*	*gap*
Frames within both residual limits	–	21%	*gap*	*gap*

in % of weight times height over OpenCap's fifteen; the knee adduction moment's early-stance peak; the vasti symmetry index in squats by its AUC against the EMG's own label; activations by on–off agreement and correlation with each EMG channel; and the pelvis residual by the share of frames within OpenSim's limits.

Held-out subjects. The only settings chosen on LabValidation are chosen on subjects 2 to 5, against their force plates and never against joint angles; subjects 7 to 11 are never used to choose anything and are reported apart. The tool that tunes MOTOR refuses held-out subjects.

3 Results

The detail sits in the appendices: the full technical validation of MOTOR in Appendix A, the warm start's in Appendix B. This section reports the headline. The whole-trial solver's validation has not run yet; every result of MOTOR below is a gap that waits on steps 4 to 6 of its build and on its full run on the 720 inputs (#98).

3.1 Matched comparison with OpenCap

OpenCap validates on the same cohort and reference standard, which makes it the most direct external comparison: from two videos it reports a joint angle MAE of 4.5° [6], and OpenCap Monocular reports 4.8° from the single 45° view on the non-jumping activities [7]. Table 1 sets MOTOR beside them, with the warm start as a baseline.

Gap: MOTOR's matched comparison: its pooled MAE and cMAE on the envelope with the per-subject bootstrap's 95% CI, beside OpenCap Monocular's 4.8° and OpenCap's 4.5°, and its vertical force per foot against the plates (a figure of the median walking and squat trial). Waits on the whole-trial solver's full run (#98).

3.2 Accuracy regimes

Gap: Which joint angles, force axes, moments and muscles MOTOR tracks well, moderately and poorly (by held-out correlation and cMAE, from Appendix A), and how they compare with the warm start's regimes, where the elbows and toes carry constant offsets from the input and the pelvis residual stays above OpenSim's limits. Waits on the full run (#98).

Table 2: Stage A of the whole-trial solve during its build, the 8 training trials (subjects 2–5, Cam1): the warm start (the chain), stage A before the contact was settled (step 2), and after with its settings tuned (step 3). "Own contact": the force stage A's contact gives; "chain on the path": the chain's force estimate run on stage A's path. GRF in % body weight, moments in % weight × height. Development measurements, not results.

	Chain	Stage A, step 2	Stage A, step 3
Trials with the residual within limits on every frame	–	5 of 8	7 of 8
Joint angle MAE (°)	7.85	9.58	8.18
GRF vertical MAE, own contact	–	28.3	21.1
GRF anterior MAE, own contact	–	13.3	9.6
GRF vertical MAE, chain on the path	16.7	21.7	15.9
GRF anterior MAE, chain on the path	5.3	6.1	4.9
Joint moments MAE	1.50	2.16	1.81

3.3 Speed

OpenCap's own recorded dynamic solves in its LabValidation release take 2635 s for a squat, 4096 s for a drop jump and 4196 s for a walk, on windows of 0.99 to 1.68 s with an 80-muscle model; the warm start takes a median of 66 s per trial on one NVIDIA L4 (Table 19).

Gap: MOTOR's wall time per trial on one GPU, the whole-trial solve included, per activity, beside OpenCap's recorded solves, and its bit-for-bit repeatability across two runs. Waits on step 5, batching (#98).

3.4 Development measurements

Stage A of the whole-trial solve has run only on eight training trials (subjects 2 to 5, the 45° camera, one L4 each), during its build; we report it as a development measurement, not as a result (Table 2). With the contact settled at the warm start, stage A holds the pelvis residual within both of OpenSim's limits on every frame of 7 of the 8 trials, and of the sit-to-stand on 98% of its frames; before the settling it held 5. On the paper's measures it is not yet a gain over its own warm start: the joint angle error rises from the chain's 7.85° to 8.18°, and the joint moments from 1.50 to 1.81% of weight times height, while the chain's own force estimate on stage A's path is close to the chain's (vertical 15.9 against 16.7% of body weight). A walk takes 45 to 90 s and a squat or sit-to-stand 6 to 12 min.

Two tries to improve the forces did not hold and were not kept. The first widened the foot's contact so that a seen stance foot pushed over its whole sole (36 training trials of the 45° camera): a wider sole let the solve move the centre of pressure wherever the movement's moment demand wanted it, so the anterior force error rose from 7.30 to 8.09% of body weight and the centred centre-of-pressure error from 51.6 to 62.7 mm as the sole widened from 5 to 40 mm. The second took the settings that scored best on the plates in step 3 (a stiffness of 2×10^7 N/m, a slip transition of 0.2 m/s, a friction of 0.6 and an effort share of 0.01): its anterior force fell from 9.6 to 6.2% of body weight and its joint moments from 1.81 to 1.66%, but the sit-to-stand's residual held on only 1% of its frames; the landmark weights learned by the EM step of [12] showed the same loss there. Both point to the same place: the pelvis' moment demand from the video's kinematics, and the seat's contact.

4 Discussion

4.1 A solver from any body

MOTOR takes a moving body, not a video. Its input is what parametric body models now produce: a skin mesh, a rotation per segment and the joint centres. This makes the same solver serve several sources. A video estimator such as GEM-X supplies it from one phone, as in the validation; a motion generator such as Kimodo supplies it from text, and SCORE runs MOTOR on Kimodo's motion to give its generated people forces and muscles; motion capture supplies it through a body model fitted to the markers; and the SOMA convention [26], which recovers one skeleton from the posed meshes of several body models, opens the route from other estimators. A newer or faster estimator can be swapped in without changing the solver.

The choice of inputs is also what makes the fit well posed. A segment rotation fixes a bone group's turn about its own long axis, which markers a few centimetres apart resolve poorly, so the rotations are the main term and the markers place the joint centres and lengths beside them. The skin gives the fit a target that a marker set does not: the bones must lie inside the body they came from, a constraint SKEL builds into its body model [13] and MOTOR takes from the input. And the fit is judged on what it is for, how well the model follows the body, rather than on the marker distances it pulls on, which a solver can always lower by trading one landmark for another.

Running every frame and every step on one GPU changes what a solve costs. The warm start already solves a trial in about a minute where OpenCap's recorded solves take 44 to 70 minutes on the CPU; the models and hardware differ, and the whole-trial solve adds its own time. Determinism matters as much: in Warp's deterministic mode two runs of the chain agree bit for bit, which answers, for this chain, the CPU against GPU drift reported on MyoSuite bodies [35].

4.2 Limitations

The solver is not yet validated. Steps 4 to 6 of the whole-trial solve (the muscles, batching and the move into the stage) are being built, and stage A has run only on eight training trials, where it does not yet improve on its warm start. Every claim about MOTOR's accuracy and speed waits on its full run.

What the input carries. MOTOR follows its input, so the input's errors pass through. In the warm start the elbows come out 21 to 24° straighter than the lab's in every envelope trial with a centred error of 3 to 6°; on 540 walking arm trials GEM-X's own elbow is a median 18.5° straighter than the markers, and the fit follows it within 2.5°. The pelvis tilt is biased by about 5° in one direction in 82 of 89 envelope trials. The lengths are SOMA's, so an error in the estimated body shape becomes an error in the model's size. Monocular depth ambiguity limits the out-of-plane rotations, as it does for every single-camera method.

Forces and residuals. In the warm start the pelvis residual stays above OpenSim's limits (medians of 7.0% of body weight and 8.9% of weight times height), the anterior ground reaction force is about twice OpenCap's, and the activation pattern is partly off (rectus femoris does not follow its EMG, $r = -0.08$). Holding the residual is what the whole-trial solve is for; whether it does so while keeping the forces and moments right is what its validation will show.

The fit's own measures. Level 1 tunes on synthetic SOMA bodies, not on people. The last run whose held-out score is recorded met its rule: the check bodies scored 3.56 against the tuning

bodies' 3.51, from a start of 4.53; the point now on main came from a later run with a broader objective, whose held-out score is still to be recorded. The bones are not yet all inside the skin: on the fifteen clips of SCORE's mean test body 5.15% of the bone points lie outside it, averaged over body regions, against a target of under 5%.

One cohort. LabValidation is nine healthy young adults in a laboratory. MOTOR holds every joint within the published healthy ranges; people whose joints move past or short of them, as many patients do, need ranges of their own, which MOTOR does not yet solve.

4.3 Relationship to adjacent work

MOTOR shares its aims with OpenCap [6, 7], from which it takes its cohort, its reference standard, its contact model's form and its residual limits; it differs in taking a body model's rotations and skin as input and in solving on the GPU. Cotton's group showed first that biomechanical models can be fitted from video in MuJoCo on the GPU [8], that one body scale can be fitted across all of a person's trials [10], that torques and forces can be recovered from markerless capture [11], and that landmark weights can be learned from the solve's own residuals [12]; MOTOR's level 2 and its tuning use the last two ideas. SKEL [13] and AddBiomechanics [14] put a biomechanical skeleton inside a body model and fit models to motion capture; MOTOR keeps the input body as the truth and fits the model inside it. MuscleMimic [19], MS-Human-700 [20] and LocoMuJoCo [36] use musculoskeletal bodies on MuJoCo and MuJoCo Warp to learn controllers; MOTOR solves for the forces of a given motion. The author's earlier work reads clinical joint angles straight from the same body model's rotations [34]; MOTOR uses none of that method and goes further, to forces and muscles.

5 Conclusion

MOTOR is an open musculoskeletal solver that takes a moving body from any source, a skin mesh, segment rotations and joint centres, and drives MyoFullBody with it: one fit per frame with the segment rotations as the main target, anatomical landmarks beside them, the bones inside the skin and every joint within its limits, solved in three levels, and then the whole trial as one dynamics-consistent optimisation on a single GPU with Newton and MuJoCo Warp. Its warm start already solves a trial in about a minute and repeats bit for bit.

Gap: The conclusion's measured sentence goes here: MOTOR's accuracy against OpenCap on the same cohort and reference, on the held-out subjects, and its time per trial. Waits on the whole-trial solver's full run (#98).

Forces and muscle activity have been out of reach wherever there is no laboratory. A solver that takes the body any modern estimator or generator produces, and returns the forces in about a minute on one card, is meant to bring them to wherever a movement can be filmed or generated.

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A Full technical validation of MOTOR

Laid out as Appendix B of [34]: all 720 single-camera inputs (every activity including drop jumps, each camera its own input), every joint angle MOTOR outputs, ground reaction forces on all three axes per activity, joint moments, residuals, every EMG channel, per activity and per camera, with the held-out subjects 7–11 apart. Every table waits on the whole-trial solver's full run (#98); Appendix B has the same tables for the warm start.

A.1 Every joint angle

Table 3: MAE, cMAE, RMSE and r per coordinate on the envelope and on all inputs, the matched 15 first.

Gap: Every coordinate MOTOR outputs (the 31 of Appendix B's table): MAE, cMAE, RMSE and mean r on the matched envelope and over all scored inputs, the dagger for a reference range under 15°, the pooled matched 15, and the held-out subjects' rows apart; the warm start beside each.

Table 4: Per coordinate and activity.

Gap: MAE and cMAE of every coordinate per activity (walking, walking with trunk sway, squats, sit-to-stand, drop jump), with cycle-aligned waveforms of the hip, knee and ankle per activity.

A.2 Per subject group, activity and camera

Table 5: Per group: inputs, matched MAE and cMAE, vertical GRF MAE, median vertical r per foot, joint moment MAE, mean EMG r.

Gap: Rows: held out 7–11, training 2–5, each of the five activities and each of the five cameras (0°, ±45°, ±70°); MOTOR's values with the warm start's in brackets.

A.3 Ground reaction force

Table 6: Ground reaction force per activity on all three axes.

Gap: MAE in % body weight, vertical, anterior and medio-lateral, both feet summed as OpenCap scores them, per activity and held out apart; the share of the plates' stance frames each foot carries; the centre-of-pressure error; OpenCap's two-camera row beside it.

A.4 Joint moments

Table 7: Joint moments per coordinate and activity.

Gap: MAE in % weight × height for each of OpenCap's fifteen joint moments per activity, the knee adduction moment's early- and late-stance peaks in walking, against OpenSim's inverse dynamics on the lab's model; OpenCap's published values beside them.

A.5 Residuals

Table 8: The pelvis residual per group.

Gap: Per group: the median force and moment, the 95th-percentile force, and the share of frames within both of OpenSim's limits (2% body weight, 1% weight × height).

A.6 Muscle activations against EMG

Table 9: Each of the 16 EMG channels against MOTOR's activation of that muscle.

Gap: On–off agreement, Pearson's r and the lag at the cross-correlation peak for each channel (biceps femoris long head, semitendinosus, gastrocnemius medialis, soleus, gluteus medius, vastus lateralis and medialis on both sides, rectus femoris and tibialis anterior on the left), per activity and held out apart; the vasti symmetry AUC per camera against the instructed and the EMG-labelled asymmetry.

A.7 Time per step and repeatability

Table 10: MOTOR's wall time per step on one GPU.

Gap: Median, range and per-frame seconds of each step (fit levels 2 and 3, the whole-trial solve's stages, the muscles) over all scored inputs, the GPU named; two runs compared bit for bit.

A.8 Failures

Table 11: The inputs without a score, by cause.

Gap: Every input of the 720 that did not run or was not scored, by cause.

B The warm start's full validation

These tables are the per-frame chain's (Section 2.7): the fit, footing, residual reduction and two-stage static optimisation, which is now the whole-trial solve's warm start. They are not MOTOR's results. In the chain's final run of 7 November 2026, all 720 single-camera inputs were prepared and sent to the cluster; 715 ran and were scored, and Section B.7 gives the rest by cause. "This run" below is that run; "before" and "the previous full run" are the chain's earlier full run of the same 720 inputs (700 scored), before its contact, ground and ligament changes, scored again with the same code. Every number is from paper/evidence/sim-body/labval-final-2026-11-07.txt.

B.1 Every joint angle

Table 12: MAE and centred MAE (degrees) per coordinate, previous full run → this run, and this run's mean r (†: the reference moves less than 15°). Envelope: 45° camera, no drop jumps (72 → 89 trials). All: every scored input (700 → 715). The matched 15 first; pro_sup has no shared zero, so only its cMAE.

	Envelope			All		
Coordinate	MAE	cMAE	r	MAE	cMAE	r
ankle angle r	6.41 → 6.39	3.18 → 3.19	0.89	5.97 → 5.96	3.94 → 3.92	0.92
ankle angle l	4.45 → 4.61	3.22 → 3.32	0.89	5.52 → 5.51	3.75 → 3.75	0.92
knee angle r	6.30 → 6.06	3.16 → 3.04	0.99	6.02 → 5.99	3.18 → 3.17	0.99
knee angle l	6.10 → 5.94	3.33 → 3.24	0.98	6.11 → 6.09	3.51 → 3.50	0.98
hip flexion r	8.17 → 8.03	3.90 → 3.82	0.98	8.91 → 8.89	4.16 → 4.11	0.98
hip flexion l	7.46 → 7.37	3.56 → 3.53	0.98	8.96 → 8.93	4.22 → 4.17	0.98
hip adduction r	4.20 → 4.20	2.81 → 2.73	0.52†	3.69 → 3.72	2.58 → 2.59	0.52†
hip adduction l	4.78 → 4.72	2.83 → 2.76	0.60†	3.48 → 3.52	2.50 → 2.52	0.57†
hip rotation r	4.10 → 4.07	2.62 → 2.62	0.43†	4.38 → 4.36	2.77 → 2.77	0.51
hip rotation l	4.53 → 4.41	3.19 → 3.06	0.43†	4.22 → 4.20	3.03 → 3.00	0.54
pelvis tilt	5.93 → 6.06	3.60 → 3.64	0.51	7.05 → 7.02	4.43 → 4.39	0.40
pelvis list	3.26 → 3.15	2.48 → 2.46	0.17†	3.10 → 3.11	2.22 → 2.24	0.16†
lumbar extension	4.66 → 4.56	3.15 → 3.11	0.64	5.36 → 5.32	3.37 → 3.35	0.75
lumbar bending	4.37 → 4.19	3.31 → 3.21	0.44	3.68 → 3.69	2.51 → 2.54	0.45†
lumbar rotation	3.67 → 3.67	2.14 → 2.12	0.58†	4.01 → 3.97	2.74 → 2.71	0.44
arm add l	3.79 → 3.98	2.33 → 2.34	0.70	6.62 → 6.58	5.18 → 5.12	0.70
arm add r	4.30 → 4.23	2.25 → 2.23	0.51†	6.22 → 6.16	4.45 → 4.40	0.64
arm flex l	7.92 → 8.03	3.11 → 3.19	0.93	12.59 → 12.46	8.72 → 8.59	0.90
arm flex r	6.77 → 7.00	3.09 → 3.19	0.93	11.32 → 11.20	8.65 → 8.52	0.90
arm rot l	8.19 → 8.01	4.41 → 4.50	0.64	12.48 → 12.35	9.91 → 9.78	0.67
arm rot r	7.25 → 7.03	4.27 → 4.09	0.67	11.31 → 11.18	8.74 → 8.61	0.68
elbow flex l	23.80 → 23.68	4.74 → 4.61	0.74	24.77 → 24.70	5.73 → 5.67	0.79
elbow flex r	20.96 → 21.12	3.07 → 3.00	0.80	22.79 → 22.74	5.84 → 5.76	0.76
mtp angle l	15.23 → 15.35	2.49 → 2.47	0.15†	15.73 → 15.74	2.19 → 2.19	0.10†
mtp angle r	15.17 → 15.22	2.58 → 2.55	-0.02†	14.99 → 14.99	2.05 → 2.06	0.00†
pelvis rotation	2.76 → 2.70	2.63 → 2.57	0.50†	2.66 → 2.63	2.45 → 2.42	0.48†
pro sup l	-	2.82 → 2.82	0.16†	-	3.63 → 3.61	0.25
pro sup r	-	2.44 → 2.55	0.18†	-	3.48 → 3.46	0.26†
subtalar angle l	9.18 → 9.24	7.02 → 6.88	-0.21	9.74 → 9.73	6.25 → 6.25	-0.07
subtalar angle r	9.02 → 9.16	6.26 → 6.26	0.04	11.44 → 11.39	6.21 → 6.21	0.04
matched 15, pooled	5.23 → 5.16	3.10 → 3.06		5.36 → 5.35	3.26 → 3.25	

The angles are the same chain's in both runs (GEM-X and the fit are unchanged); the small differences come from the inputs each run has. The elbows are about 21 to 25° off with a centred

error of 3 to 6° (straighter than the lab's in each of the 89 envelope trials; GEM-X's arm estimate, which the fit follows), and the toes (mtp) 15° off with a centred 2 to 3°: constant offsets, not tracking errors.

B.2 Per subject group, activity and camera

Table 13: This run per group (the previous full run's value in brackets): inputs, matched MAE and cMAE (degrees), vertical GRF MAE (% body weight, both feet), median vertical r per foot, joint moment MAE over OpenCap's 15 (% weight × height), mean EMG r over the channels. Cameras: LabValidation's numbering; Cam1 is the envelope's 45° camera.

Group	n	MAE	cMAE	GRF vert.	r R / L	Moments	EMG r
held out 7–11	396	5.13 (5.14)	3.02 (3.02)	16.28 (54.33)	0.77 / 0.79	2.41 (2.43)	0.24
training 2–5	319	5.62 (5.65)	3.54 (3.56)	16.15 (53.07)	0.78 / 0.77	2.42 (2.48)	0.22
walking	135	4.38 (4.40)	2.92 (2.93)	19.55 (76.11)	0.82 / 0.80	1.88 (1.79)	0.27
walking (trunk sway)	135	4.67 (4.67)	3.05 (3.05)	16.31 (84.14)	0.82 / 0.84	1.72 (1.53)	0.27
squats	87	5.79 (5.79)	3.26 (3.28)	6.32 (47.18)	0.53 / 0.47	1.58 (1.54)	0.17
sit-to-stand	90	5.39 (5.34)	2.92 (2.91)	11.58 (47.71)	0.68 / 0.80	1.55 (1.74)	0.20
drop jump	268	6.03 (6.03)	3.63 (3.63)	19.27 (32.21)	0.78 / 0.78	3.59 (3.73)	0.22
Cam0	144	5.21 (5.20)	3.14 (3.14)	14.95 (54.10)	0.75 / 0.83	2.44 (2.44)	0.23
Cam1 (45°)	141	5.62 (5.72)	3.33 (3.40)	15.50 (53.27)	0.73 / 0.81	2.54 (2.65)	0.22
Cam2	144	5.36 (5.36)	3.53 (3.53)	15.54 (52.72)	0.81 / 0.82	2.21 (2.31)	0.25
Cam3	143	5.35 (5.35)	3.20 (3.21)	15.96 (53.03)	0.81 / 0.70	2.44 (2.44)	0.23
Cam4	143	5.23 (5.23)	3.05 (3.05)	19.16 (55.68)	0.78 / 0.75	2.43 (2.45)	0.23

B.3 Ground reaction force

Table 14: Ground reaction force per activity, this run (previous run in brackets): MAE in % body weight on three axes with both feet summed, as OpenCap scored it; the share of the plate's stance frames the run's foot carries.

Activity	vertical	anterior	medio-lateral	3-axis mean	stance carried R / L
all	16.22 (53.77)	7.41 (6.45)	2.99 (3.19)	8.9 (21.1)	0.82 / 0.79
walking	19.55 (76.11)	12.03 (9.65)	3.02 (3.35)	11.5 (29.7)	0.82 / 0.75
walking (trunk sway)	16.31 (84.14)	10.86 (9.11)	3.49 (4.21)	10.2 (32.5)	0.83 / 0.85
squats	6.32 (47.18)	3.53 (2.56)	3.30 (4.17)	4.4 (18.0)	0.99 / 0.95
sit-to-stand	11.58 (47.71)	4.42 (4.74)	3.25 (3.51)	6.4 (18.7)	0.82 / 0.82
drop jump	19.27 (32.21)	5.61 (5.37)	2.54 (2.20)	9.1 (13.3)	0.76 / 0.71
OpenCap, 2 cameras	11.4	3.5	3.8	6.2	–

B.4 Residuals

Table 15: The pelvis residual this run leaves, per group: the median over trials of each trial's median force (% body weight) and moment (% weight × centre-of-mass height), the median of each trial's 95th-percentile force, and the mean share of frames within both of OpenSim's limits (2% and 1%). No trial is within them on every frame.

Group	trials	force	moment	force, 95th pct.	frames within
all	715	7.0	8.90	24.4	21%
held out 7–11	396	7.1	9.31	23.9	21%
walking	135	7.8	10.99	23.0	6%
walking (trunk sway)	135	7.0	9.59	20.3	7%
squats	87	0.7	1.06	8.4	49%
sit-to-stand	90	0.2	2.82	8.8	46%
drop jump	268	10.9	9.32	85.5	17%

B.5 Muscle activations against EMG

Table 16: Each EMG channel against the chain's activation of that muscle, all 715 inputs: the share of frames on which both are on or both off, Pearson's r, and the lag (s) at the cross-correlation peak.

Muscle	on/off	r	lag	Muscle	on/off	r	lag
bflh l	0.65	0.13	-0.130	semiten r	0.70	0.23	-0.034
bflh r	0.65	0.13	-0.144	soleus l	0.70	0.26	-0.030
gasmed l	0.70	0.23	-0.009	soleus r	0.69	0.28	-0.041
gasmed r	0.69	0.26	-0.022	tibant l	0.59	0.22	+0.031
glmed1 l	0.51	0.13	+0.051	vaslat l	0.72	0.40	-0.003
glmed1 r	0.52	0.15	+0.040	vaslat r	0.72	0.42	-0.018
recfem l	0.56	-0.08	-0.002	vasmed l	0.72	0.36	-0.024
semiten l	0.66	0.18	-0.053	vasmed r	0.76	0.41	-0.033

Table 17: OpenCap's vasti symmetry index in squats: the AUC of the chain's index against the asymmetry the instructions asked for and against the EMG's own label (index over 1.15), previous run → this run. OpenCap: 0.83 against the EMG label.

Camera	AUC, instructed	AUC, EMG label
Cam0	0.47 → 0.84	0.64 → 0.75
Cam1	0.52 → 0.72	0.08 → 0.62
Cam2	0.60 → 0.95	0.52 → 0.84
Cam3	0.75 → 0.90	0.65 → 0.83
Cam4	0.53 → 0.81	0.25 → 0.48
all	0.56 → 0.73	0.45 → 0.66

B.6 Time per step

Table 18: The chain's wall seconds per step on one NVIDIA L4, over the 715 runs: median, range, and the median per frame (clips of 30 Hz).

Step	median s	min s	max s	s per frame
fit	21.825	14.483	623.317	0.303
set up: Newton's model, its card, the filtered path	4.697	3.878	32.851	0.061
footing	4.669	1.758	35.312	0.062
RRA, rounds	19.862	5.910	392.306	0.246
RRA, follows and frame caps	4.603	1.714	45.888	0.061
checks: RRA's record, feet held, following on MuJoCo's compile	2.120	1.111	17.879	0.028
static optimisation 1, ground reaction	3.003	0.114	28.318	0.034
static optimisation 2, stops and muscles	2.296	1.169	22.356	0.032
inverse dynamics, contacts and muscle paths	1.120	0.356	12.459	0.013
records and push measures	0.264	0.042	2.639	0.003
arrays	0.005	0.003	0.012	0.000
total	66.214	34.843	788.668	0.871

Table 19: Wall time per trial, the chain against published solves. OpenCap's solves are from its LabValidation release (80 muscles, windows of 0.99 to 1.68 s, solve only, hardware not recorded).

Chain	Clip	Wall time
OpenCap solve [6]	walk, drop jump, squat	4196, 4096, 2635 s
Moco, MocoTrack [4]	gait	about 130 min
OpenSim CMC [3]	gait	about 10 min
Warm start (chain), one L4	LabValidation, 715 trials	66 s median (35–789 s)

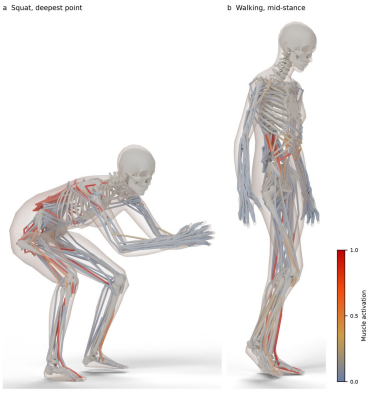


Figure 2: The warm start's MyoFullBody inside the person's SOMA skin, two LabValidation trials drawn from the chain's own saved results: (a) a squat at its deepest point, (b) walking at mid-stance; each muscle coloured by its activation.

B.7 Failures

Table 20: The 5 of 720 inputs without a score in the chain's run, by cause.

Cause	Inputs
The card's fit solve did not stop in 3000 steps (subject 10, squats1, two cameras)	2
A frame's fit did not converge in 300 steps (two drop jumps)	2
Passed its job's time limit (subject 10, squats1, a third camera)	1
Total	5

The run first stopped at its EUR 25 spend limit with 3 of its 48 jobs unfinished; their 45 inputs ran after with the same code.

Two faults found in the first scoring of an earlier run are fixed and reported because they moved the headline number then. First, 148 of the 720 GEM-X results came from another video: a resumed cloud run named its jobs by position in a shortened list, so a job picked up an earlier job's finished output; jobs are now named by a hash of their own description. Second, the time shift could not reach a walking trial's true lag: requiring the shifted reference to cover 80% of a clip about one gait cycle long capped the shift near ± 0.27 s, against the protocol's ± 1.0 s; at 50% the feet are in phase at the found lag in every walking trial. The pooled matched MAE went from 8.59° (78 trials) to 6.02° (62 trials) then.

C Disclosure

The code of MOTOR and of its validation was written by a coding agent (Claude Code) under the author's direction and review; this draft was written the same way. No result was hand-picked. The only settings chosen on LabValidation (the chain's three contact settings and box clearance, and the whole-trial solve's contact settings and effort share) were chosen on subjects 2 to 5 against their force plates, never against joint angles; subjects 7 to 11 are held out and reported apart.